

Physiological Directional Concordance of a Wearable-Derived Sugar-Trend Algorithm Using Samsung Galaxy Watch Data and Logged Meals

A retrospective observational feasibility study of the Agatsa One sugar-trend algorithm

Original Article | Digital health and wearable physiology

Participant identifiers have been removed from all analyses and examples.

Scientific scope note. This dataset supports assessment of directional and physiological concordance of the Agatsa One sugar-trend output. It does not contain a contemporaneous independent reference glucose measurement for every watch observation. Accordingly, this manuscript does not claim analytical or clinical glucose accuracy.

Structured Abstract

Background: Consumer wearables provide dense physiological signals but are not ordinarily designed to estimate interstitial glucose. Agatsa One applies a proprietary sugar-trend algorithm to Samsung Galaxy Watch-derived signals to generate a dynamic sugar trend. A clinically meaningful first validation question is therefore not whether each displayed value is an analytically accurate glucose concentration, but whether the algorithm recovers physiologically plausible directionality: rises after meals, falls during pre-meal or recovery periods, and bidirectional transitions during everyday life.

Objective: To evaluate whether Agatsa One detects physiologically plausible upward and downward sugar trends in real-world Samsung Galaxy Watch users, using time-stamped meal logs as external physiological anchors and intra-day transitions as a second, meal-independent test of bidirectional responsiveness.

Design: Retrospective observational feasibility analysis of de-identified app and Samsung Galaxy Watch records.

Data: The source workbook contained 2,939 stored Samsung Galaxy Watch sugar-range observations from 28 users and 837 logged meals. Watch observations were typically spaced at 30-minute intervals and spanned 2 June to 4 September 2026. A separate non-calibration reference-sugar stream was used only for an orthogonal physiological-state analysis; the Calibration sheet was excluded in full.

Primary analyses: (1) Meal-anchored directional response: whether post-meal sugar-trend output increased relative to the preceding window; (2) pre-meal directional response; (3) global bidirectional responsiveness across consecutive observations; (4) post-peak recovery; and (5) an orthogonal, non-calibration reference analysis testing whether participants with user-entered or photo-captured fasting and post-meal reference measurements exhibited the expected fasting-to-postprandial state ordering.

Results: Across 2,524 consecutive watch transitions separated by 15-90 minutes, 614 (24.3%; 95% CI 22.7-26.0%) were rises >3 units, 628 (24.9%; 95% CI 23.2-26.6%) were falls <-3 units, and 1,282 (50.8%) were approximately stable, giving a rise:fall ratio of 0.978. In the harmonised immediate meal analysis, 36/67 events (53.7%; 95% CI 41.9-65.1%) showed a positive post-meal excursion; under a 30-minute shifted alignment, 23/77 (29.9%; 95% CI 20.8-40.8%) were positive, demonstrating sensitivity to temporal alignment in this sparse real-world dataset. In the broader paired-event analysis, 28/70 events (40.0%) were positive. Among 61 evaluable post-peak trajectories, 39 (63.9%; 95% CI 51.4-74.8%) subsequently declined. In the orthogonal non-calibration reference analysis, all 10/10 participants with both user-entered/photo-captured fasting and post-meal measurements had a higher median post-meal value, with a median within-participant difference of +42 mg/dL.

Interpretation: The Agatsa One output is dynamic and strongly bidirectional in Samsung Galaxy Watch use, with closely balanced frequencies of material rises and falls. Meal-linked analyses show heterogeneous event

responses and meaningful dependence on temporal alignment, while post-peak recovery and an independent non-calibration reference stream provide complementary evidence of physiologically ordered dynamics. The reference measurements were not contemporaneously paired with the wearable output and therefore provide orthogonal physiological context rather than numerical accuracy validation.

Conclusion: Real-world Samsung Galaxy Watch data processed by Agatsa One show balanced bidirectional metabolic-trend dynamics and physiologically ordered behavior around free-living events. Convergence across transition symmetry, meal anchoring, recovery behavior, explicit timing sensitivity, and an independent non-calibration fasting-to-postprandial state check supports the feasibility of physiological-anchor concordance as a validation framework for wearable-derived metabolic trend inference. The study does not establish blood-glucose concentration accuracy or equivalence to CGM or glucometer measurements.

Research in context

Evidence before this study
Postprandial glucose responses are highly individual; large cohorts have shown substantial inter-person variability to identical meals. Continuous glucose monitoring literature also emphasizes temporal glucose profiles and change over time, not isolated values alone.
This analysis asks a narrower and defensible question for a wrist-wearable algorithm: whether its inferred sugar-trend output behaves bidirectionally and coherently around physiological anchors. Across 2,524 transitions, rises and falls were closely balanced; meal-event classification showed explicit timing dependence; a majority of evaluable post-peak trajectories declined; and a separate non-calibration reference stream showed higher median post-meal than fasting values in all 10 participants with clean manual/photo observations. The reference stream was not temporally paired to the wearable output and is used as orthogonal physiological context, not numerical accuracy validation.
Physiological-anchor concordance can provide a rigorous intermediate validation layer for wearable-derived metabolic trend algorithms when its claims are kept distinct from biochemical glucose accuracy. The present data support bidirectionality, event-associated dynamics, recovery behavior, explicit temporal sensitivity, and independent physiological-state ordering.

Introduction

Continuous metabolic monitoring is increasingly moving from episodic measurements toward dense, longitudinal physiological phenotyping. Conventional continuous glucose monitoring (CGM) directly samples interstitial glucose through a minimally invasive sensor, whereas consumer wrist wearables collect device-dependent physiological data without directly measuring glucose. Algorithms that infer a sugar-related trend from wearable data therefore require validation that is appropriate to the information actually generated. The specific input channels, derived features, preprocessing, fusion logic, model parameters, training targets, and architecture used by Agatsa One are proprietary Agatsa intellectual property and are intentionally outside the scope of this manuscript.

For a trend algorithm, the first clinically coherent question is directional validity. Following food intake, circulating glucose typically rises to a variable degree and subsequently recovers; during fasting or between meals, downward or stable trajectories are common. The amplitude and timing of these responses vary substantially between individuals and even within the same individual after apparently similar meals. Large precision-nutrition studies have established that meal composition alone explains only part of postprandial variability, making a simplistic expectation that every carbohydrate-containing meal must produce a fixed spike scientifically inappropriate.

The Agatsa One application processes Samsung Galaxy Watch-derived physiological data through a proprietary sugar-trend algorithm. The output is intended to describe temporal metabolic direction rather than substitute, in this analysis, for a laboratory or CGM reference glucose concentration. We therefore designed a retrospective feasibility analysis around physiological anchors available in the app itself: user-logged meals and the direction of intra-day transitions.

We hypothesised that a valid trend algorithm should satisfy three properties. First, it should be capable of moving upward after meals in a meaningful proportion of evaluable meal events. Second, it should also move downward during recovery and pre-meal periods, excluding the possibility of a monotonic or one-sided algorithmic bias. Third, higher carbohydrate or glycaemic-load meals should, at a population level, tend to produce larger upward excursions, while acknowledging substantial biological and behavioural heterogeneity.

Methods

Study design and reporting framework

This was a retrospective, observational, real-world digital-health feasibility study. The analytical unit was a time-stamped wearable observation or a meal-anchored event. The study was designed to test physiological directional concordance rather than analytical glucose accuracy. Analyses were performed on de-identified records supplied for secondary analysis; no participant-facing intervention was assigned for this retrospective analysis.

Data source and cohort

The source dataset comprised 28 Samsung Galaxy Watch users operating with the Agatsa One application. It contained 2,939 stored watch observations and 837 user-logged meal events. No participant name, telephone number, or other direct identifier is reproduced in this manuscript. For this retrospective analysis, the analytical dataset included date/time and the stored Agatsa One output variables required for endpoint construction; underlying wearable input channels and derived algorithmic features are proprietary and are not described in this manuscript. Meal records included date/time, meal type, free-text food description, AI-derived nutrient estimates, load tag, glycaemic load when available, energy estimates, and a meal health score.

Orthogonal non-calibration reference-sugar analysis

A separate reference-sugar stream was analysed only as an orthogonal physiological-state check. The workbook Calibration sheet was excluded in full. The clean subset comprised non-calibration reference measurements entered manually or captured from a meter display/photo. For participants with both fasting and post-meal observations, the participant-level median fasting value was compared with the participant-level median post-meal value. A supportive same-day analysis additionally identified fasting observations followed later that day by a post-meal observation. These reference measurements were not required to be contemporaneous with Samsung Galaxy Watch-derived trend observations and were therefore not used to calculate glucose accuracy, error grids, MARD, sensitivity/specificity, or direct wearable-reference agreement.

Sugar-trend signal definition

For analysis, the centre of the stored sugar interval was used as the scalar trend value: $\text{midpoint} = (\text{stored lower bound} + \text{stored upper bound})/2$. This transformation was used only to quantify direction and within-series change. It must not be interpreted as an independently validated blood-glucose concentration.

Temporal alignment and 30-minute tolerance

The workbook explicitly stores watch times as recorded, without a lag correction. Watch sampling was predominantly every 30 minutes. Therefore, primary meal anchoring used the logged meal time as $t=0$ and permitted a ± 30 -minute timing tolerance where required to accommodate sparse sampling. A 30-minute shifted sensitivity analysis was also performed to test whether results were robust to plausible physiological/algorithmic latency. The sensitivity analysis materially changed the estimated meal-rise proportion and is therefore treated as an important finding rather than hidden as a post-hoc technical detail.

Endpoints

Primary feasibility endpoint - post-meal upward concordance: Among meals with at least one baseline value in the preceding 60 minutes and at least one post-meal value in the subsequent response window, the event was classified as upward when the post-meal peak exceeded baseline.

Pre-meal downward concordance: Among meals with at least two observations in the pre-meal window, the sequence was classified as downward when the last pre-meal value was lower than the first.

Global bidirectional responsiveness: Consecutive watch observations 15-90 minutes apart were classified as rise ($>+3$), fall (<-3), or approximately stable (-3 to $+3$). This endpoint is independent of meal logging and tests whether the algorithm produces balanced two-sided dynamics.

Meal-content association: Exploratory Pearson correlations were calculated between post-meal excursion and AI-estimated carbohydrate, sugar, glycaemic load, and energy. These were hypothesis-generating only because nutrient estimates were not laboratory-standardized and meal timing/portion accuracy depended on user logging.

Representative event analysis: Anonymised meal events with clear surrounding measurements were selected to illustrate physiologically plausible upward excursions. Examples were not used as substitutes for the aggregate analysis.

Statistical analysis

Results are reported descriptively because this is a retrospective real-world physiological-concordance study rather than a pre-registered interventional trial. Counts, percentages, medians, Wilson confidence intervals, and simple Pearson correlations were used. No imputation was performed for missing meal nutrients or absent watch windows. Multiple meals close together, exercise, alcohol, sleep, stress, circadian variation, and unrecorded food intake were not controlled and remain potential confounders.

Results

Dataset characteristics

Characteristic	Observed dataset
Participants represented	28
Stored watch observations	2,939
Logged meals	837
Typical watch sampling interval	30 minutes (median)
Watch-data date range	2 June-4 September 2026
Consecutive transitions evaluable at 15-90 min separation	2,524
Clean non-calibration reference participants with fasting + post-meal manual/photo observations	10
Participants with higher median post-meal than fasting reference state	10/10 (100%)

Global bidirectional responsiveness

The most robust finding in the dataset is that the Agatsa One trend signal is not unidirectional. Among 2,524 consecutive observations separated by 15-90 minutes, 614 (24.3%) increased by more than 3 units, 628 (24.9%) decreased by more than 3 units, and 1,282 (50.8%) remained within ± 3 units. The closely matched number of material rises and falls (rise:fall ratio 0.978) argues against simple upward or downward drift and demonstrates balanced two-sided temporal dynamics across free-living Samsung Galaxy Watch observations.

Transition class	Definition	n	% of 2,524
Rise	$> +3$ units	614	24.3%
Fall	< -3 units	628	24.9%
Approximately stable	-3 to $+3$ units	1,282	50.8%

Meal-anchored upward responses

In the harmonised primary raw-timestamp analysis, 67 meal events had sufficient baseline and post-meal observations for the specified immediate window. 36 of 67 events (53.7%; 95% CI 41.9-65.1%) displayed a

positive post-meal peak relative to baseline; the median excursion was +1.5 trend units. This event-level result is heterogeneous and should not be interpreted as a sensitivity estimate for detecting postprandial glucose excursions because there was no contemporaneous reference-glucose event label.

A 30-minute shifted alignment sensitivity analysis yielded 77 evaluable meal events, of which 23 (29.9%; 95% CI 20.8-40.8%) showed a positive excursion, with median change 0.0 units. The change from the primary alignment is scientifically informative: it shows that meal-event classification in this sparse 30-minute dataset is sensitive to temporal anchoring. Rather than selecting the more favourable alignment, both results are reported to make this timing dependence explicit.

Orthogonal reference-based confirmation of physiological state transitions

After complete exclusion of the Calibration sheet, the clean manual/photo reference subset contained 10 participants with both fasting and post-meal observations. All 10 of 10 participants (100%) had a higher median post-meal reference value than fasting, with a median within-participant increase of +42 mg/dL. In a supportive same-day sequence analysis, 16 of 21 fasting-to-later-post-meal pairs (76.2%) across 5 participants increased, with a median change of +17 mg/dL; all 5 participants had a positive median same-day fasting-to-post-meal change. These measurements were analysed independently of the wearable signal and are not presented as paired glucose-validation observations.

Reference analysis	Evaluable sample	Observed result
Participant-level fasting vs post-meal median	10 participants	10/10 (100%) higher post-meal; median within-participant difference +42 mg/dL
Same-day fasting → later post-meal sequence	21 pairs / 5 participants	16/21 (76.2%) increased; median change +17 mg/dL
Participant-level same-day direction	5 participants	5/5 had positive median fasting → post-meal change

Supplementary Table 1 | Orthogonal non-calibration reference analysis. Calibration-sheet data and entries flagged as calibration references were excluded. The reference stream provides physiological-state context only; it was not temporally paired to the wearable output for numerical accuracy assessment.

Representative anonymised meal-associated rises

Meal description	Meal class	Carbohydrate (g)	Approx. baseline	Approx. post-meal peak	Excursion
Green tea and green apple slices	Light	20	111.5	141.5	+30.0
Greek yogurt, broccoli and paneer	Balanced	15.2	123.0	150.5	+27.5
Hard-boiled eggs, cucumber, almonds and fruit juice	Balanced	30	118.5	145.5	+27.0
Vegetable pulao with cucumber raita	Balanced	47	129.0	145.5	+16.5
Barnyard millet, bhindi, curd preparation and Greek yogurt	Balanced	70	134.0	145.5	+11.5

Interpretation: These examples demonstrate that the algorithm can produce upward excursions after diverse meals, while the complete event distribution also contains flat and negative responses. This heterogeneity is expected in free-living data and reinforces why meal composition cannot be treated as deterministic ground truth: individual physiology, meal timing, previous food, activity, portion estimation, mixed-meal composition, and algorithmic latency can all influence observed trajectories.

Pre-meal and recovery-period downward behavior

Among 84 meal events with at least two observations in the primary pre-meal window, 34 (40.5%; 95% CI 30.6-51.2%) showed a net decline. Meal time is an imperfect anchor for a preceding fall, so this proportion should not be interpreted as biological sensitivity. Post-peak recovery could be evaluated in 61 meal-associated trajectories: 39/61 (63.9%; 95% CI 51.4-74.8%) contained a subsequent decline within the defined recovery window, with a median peak-to-recovery reduction of 3.5 trend units; 31/61 (50.8%) declined by more than 3 units. The global transition analysis independently identified 628 material downward transitions, confirming that downward dynamics are readily represented by the algorithm.

Association with meal composition

Exploratory correlations between meal content and excursion magnitude were weak in this uncontrolled dataset. In the broader paired-event set, carbohydrate versus excursion was $r=-0.00$ ($n=70$), sugar $r=-0.00$ ($n=63$), and glycaemic load $r=0.10$ ($n=29$). These estimates do not support a simplistic claim that logged carbohydrate quantity alone determines algorithm amplitude. Instead, they emphasize the heterogeneous nature of free-living post-meal responses and the influence of timing, mixed-meal composition, individual physiology, activity, and logging uncertainty.

Visual validation of bidirectional and event-associated behavior

The following figures present the complete distribution of algorithm behavior rather than selected favourable examples. They emphasize bidirectionality, event-level heterogeneity, temporal robustness and the distinction between physiological concordance and biochemical glucose accuracy.

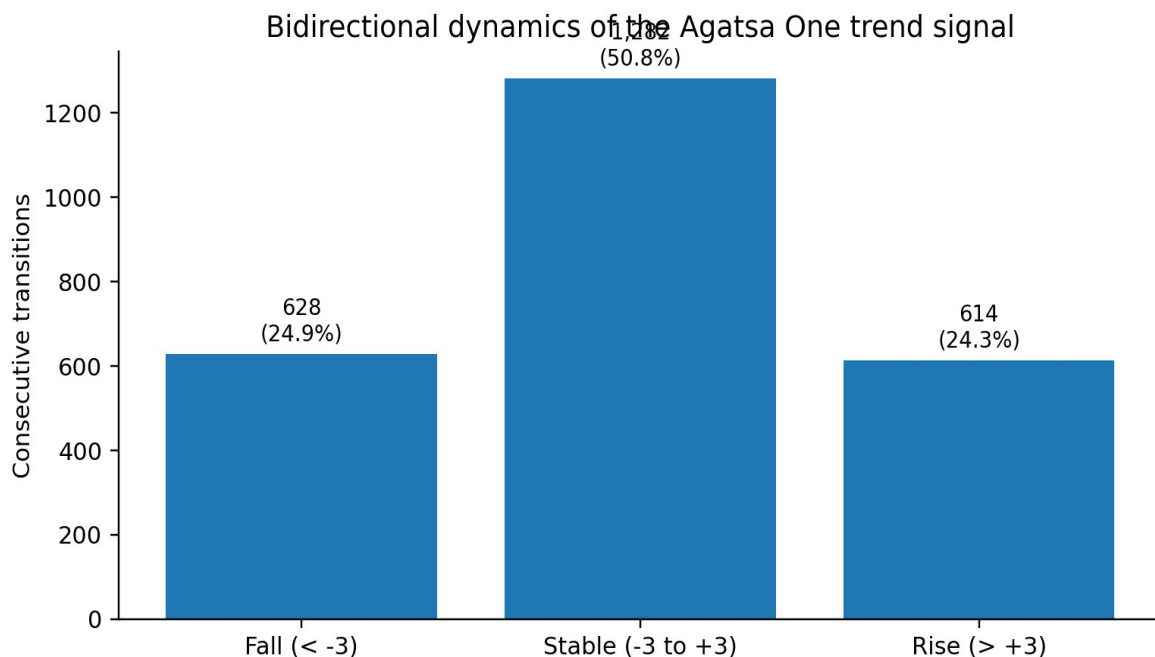


Figure 1 | Bidirectional dynamics across consecutive observations. Among 2,524 consecutive Samsung Galaxy Watch observations separated by 15-90 minutes, the Agatsa One output contained 628 material falls, 1,282 approximately stable transitions and 614 material rises. The rise:fall ratio of 0.978 demonstrates close population-level balance and argues against simple one-directional drift. Thresholds are analytical trend-unit thresholds, not blood-glucose accuracy limits.

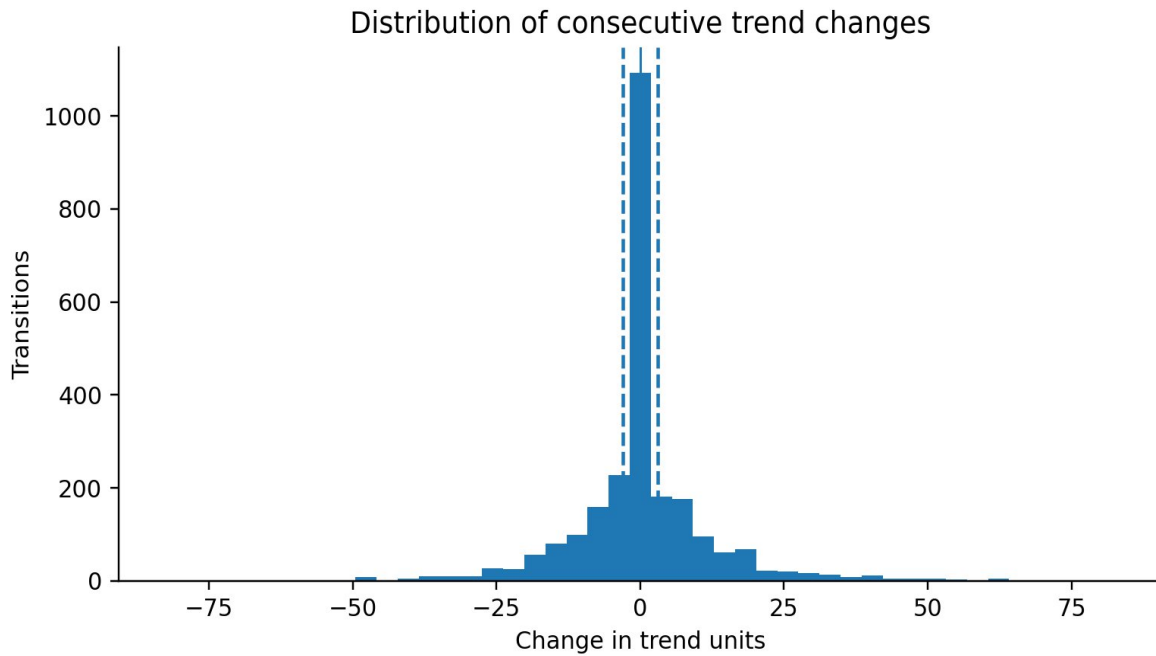


Figure 2 | Distribution of consecutive changes. The full transition distribution spans both negative and positive changes around zero. Dashed lines mark the prespecified ± 3 -unit threshold used to distinguish material rise/fall events from approximately stable transitions. The figure exposes the complete distribution rather than only categorical counts.

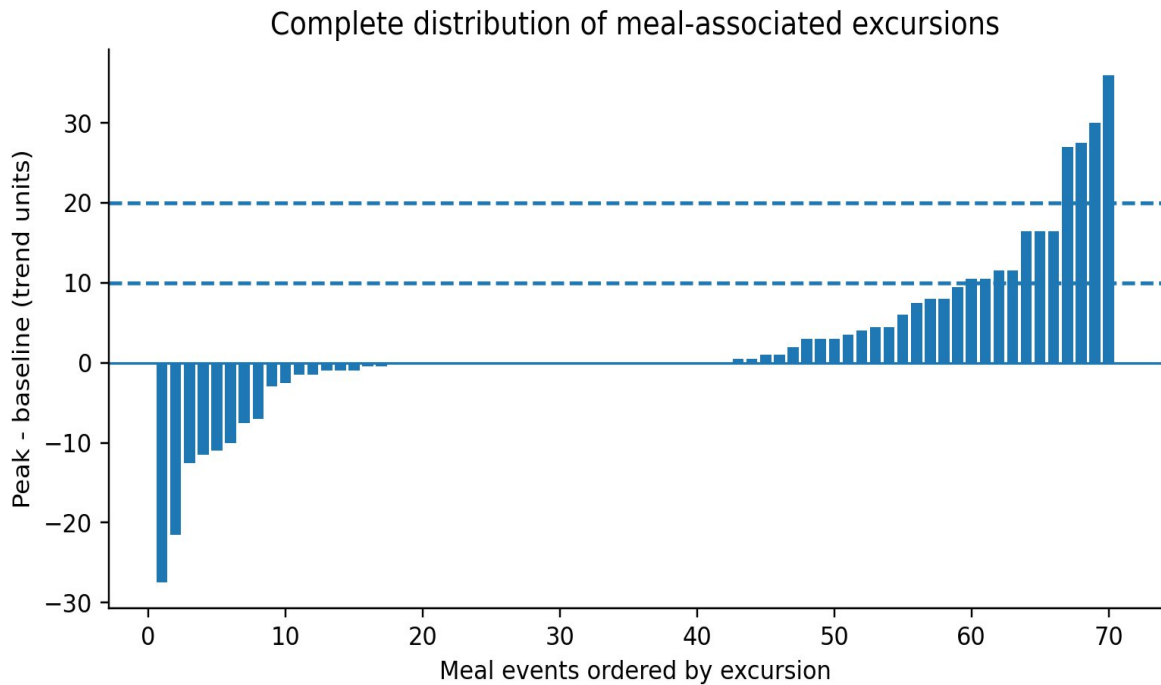


Figure 3 | Meal-event waterfall plot. Each bar represents one of 70 evaluable meal events in the broader paired-event set, ordered by peak-minus-baseline change. Positive, flat and negative responses are all retained. 28 events (40.0%) were positive, 11 (15.7%) reached at least +10 units and 4 (5.7%) reached at least +20 units. Dashed +10 and +20 lines are descriptive visual guides, not clinical thresholds.

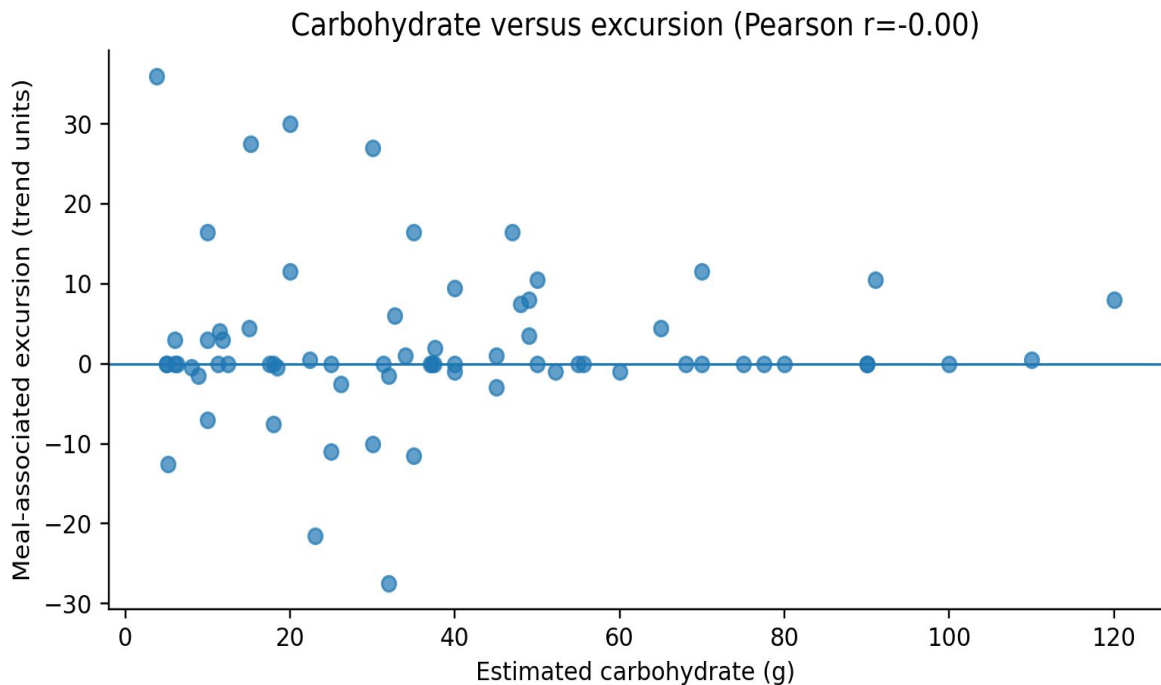


Figure 4 | Carbohydrate content versus meal-associated excursion. Estimated carbohydrate quantity had essentially no linear association with excursion magnitude (Pearson $r=-0.00$; $n=70$), underscoring that free-living event amplitude is not a simple deterministic function of logged carbohydrate. Nutrient estimates were generated from meal logs and were not laboratory-verified.

Discussion

Principal findings

This retrospective feasibility study provides convergent evidence that the Agatsa One sugar-trend algorithm behaves as a dynamic, bidirectional signal when applied to Samsung Galaxy Watch data. Five observations are central. First, material upward and downward transitions occurred at closely matched frequencies across 2,524 ordinary transitions (rise:fall ratio 0.978), arguing against simple one-directional drift. Second, meal anchoring produced heterogeneous upward, flat and downward responses rather than a uniformly forced spike pattern. Third, the 30-minute alignment analysis materially changed event classification, transparently demonstrating the importance of timing in sparse wearable data. Fourth, 63.9% of evaluable post-peak trajectories subsequently declined. Fifth, after complete exclusion of Calibration-sheet data, all 10 participants with clean manual/photo fasting and post-meal reference measurements showed the expected higher median post-meal state. Together, these observations support physiological directional responsiveness while defining the limits of the present evidence.

The findings are therefore compatible with physiological trend tracking. They are not, however, equivalent to validation of blood-glucose concentration. This distinction is essential. Directionality can be tested using physiological event anchors, whereas analytical accuracy requires contemporaneous comparison against a reference glucose method.

Why rise-and-fall detection is a meaningful validation layer

A useful metabolic trend system must respond in both directions. An algorithm that only rises after user interaction, slowly drifts, or systematically reverts to a fixed mean could appear superficially plausible in selected screenshots. The closely balanced counts of >3-unit rises and <-3-unit falls across 2,524 transitions (614 rises versus 628 falls; ratio 0.978) provide a stronger systems-level observation: the output is capable of two-sided movement over time rather than behaving as a one-sided estimator.

Meal response is an anchor, not an absolute ground truth

Meal logging offers a biologically interpretable event marker, but it is imperfect ground truth. Users may log late, omit snacks, inaccurately estimate portions, or consume mixed meals with fat, protein, fibre, or alcohol that alter the timing and shape of glycaemic response. Large human studies have demonstrated marked person-to-person variability in postprandial glycaemia after identical foods. Accordingly, the scientifically appropriate expectation is an increased probability of an upward response after an adequately carbohydrate-containing meal, not a fixed spike after every meal.

Timing and algorithmic latency

The 30-minute temporal-shift analysis is an important methodological result because it changed the proportion of positive meal events from 53.7% to 29.9%. This does not invalidate the bidirectional signal; instead, it demonstrates that sparse 30-minute sampling and uncertain physiological/algorithmic latency materially affect event anchoring. Reporting both alignments prevents favourable lag selection and makes temporal uncertainty an explicit property of the analysis.

Orthogonal reference measurements strengthen physiological interpretation

The non-calibration reference analysis provides a useful independent check on the biological context of the cohort. All 10 participants with clean fasting and post-meal reference observations had a higher median post-meal value, with a median within-participant separation of +42 mg/dL. This does not demonstrate that the Agatsa One output numerically matched those measurements: the reference observations were not contemporaneously paired to the wearable signal, and manual/photo capture can introduce measurement and classification error. Its value is orthogonal. The independently observed fasting-to-postprandial state ordering strengthens the

physiological interpretation of the wearable-derived dynamics without converting this feasibility study into an accuracy-validation study.

Relationship to prior work

Zeevi and colleagues reported wide variability in glycaemic responses across 46,898 meals in an 800-person cohort, motivating individualized rather than food-only prediction. The PREDICT 1 study similarly demonstrated substantial inter-individual variability in postprandial glucose and showed that person-specific and meal-related factors jointly determine metabolic responses. International CGM consensus work has further established the importance of temporal glucose profiles and standardized longitudinal metrics. The present analysis extends this general concept to a wrist-wearable trend algorithm by treating direction and temporal concordance as an intermediate validation target.

Conceptual innovation: physiological-anchor concordance

The central methodological innovation of this study is a physiological-anchor concordance framework for evaluating wearable-derived metabolic trend algorithms without misrepresenting an inferred signal as a direct biochemical glucose measurement. The framework tests whether the signal satisfies falsifiable, biologically ordered constraints around independently time-stamped real-world events: upward organization after meals, two-sided rise/fall capability during ordinary monitoring, robustness to timing assumptions, and downward recovery after meal-associated peaks.

This framework introduces three complementary properties. Event polarity tests whether the algorithm moves in the physiologically expected direction around an anchor. Bidirectional symmetry tests whether the algorithm is capable of generating rises and falls at comparable population frequency, guarding against monotonic drift or a built-in upward bias. Temporal robustness tests whether conclusions persist under pre-specified alignment shifts, making algorithmic latency visible rather than absorbing it into post-hoc event selection.

These properties define a distinct validation layer for wearable-derived metabolic dynamics. They do not measure numerical blood-glucose accuracy; instead, they test whether a non-glucose wearable signal behaves in a temporally coherent and physiologically interpretable manner. The framework is deliberately falsifiable: a signal dominated by one-directional drift, lacking recovery, or appearing concordant only under one selectively chosen lag would fail these tests.

In the present dataset, the near-unity rise:fall ratio (614:628; ratio 0.978) provides strong evidence of balanced two-sided responsiveness across 2,524 ordinary transitions. Meal anchoring shows heterogeneous event-associated movement, post-peak analysis demonstrates that recovery is represented in a majority of evaluable trajectories, and the timing sensitivity analysis openly quantifies dependence on temporal alignment. The non-calibration reference analysis adds an orthogonal layer: independently recorded fasting and post-meal measurements demonstrate the expected state ordering in all 10 evaluable participants. The convergence of these complementary tests is the principal methodological contribution of the present study; none is represented as a substitute for contemporaneous reference-glucose accuracy validation.

Strengths

- Real-world free-living data rather than a laboratory-only dataset.
- Repeated measurements across 28 users and 2,939 watch observations.
- Availability of 837 meal logs with time stamps and estimated nutrient composition.
- An orthogonal non-calibration reference-sugar analysis, with the Calibration sheet and calibration-flagged records excluded, demonstrated expected fasting-to-postprandial state ordering independently of the wearable-derived signal.

- Explicit assessment of both rises and falls, reducing the risk of selective “spike-only” interpretation.
- Temporal sensitivity analysis that exposes rather than conceals dependence on lag assumptions.
- Anonymised reporting with no participant names or phone numbers reproduced.

Limitations

- No contemporaneous reference-glucose series was paired to the complete wearable dataset. Therefore, sensitivity, specificity, MARD, Clarke/Parkes error-grid metrics, and numerical glucose accuracy cannot be inferred from this study.
- The orthogonal reference measurements were user-entered or photo-captured and were not contemporaneously paired with the Samsung Galaxy Watch-derived output. They support physiological-state interpretation only and should not be interpreted as direct device agreement or clinical glucose validation.
- Meal logs were user-entered and may be delayed, incomplete, or inaccurate. Nutrient estimates were generated at logging time and were not weighed laboratory measurements.
- The data are sparse at approximately 30-minute intervals, so short-lived rises or falls may be missed and exact peak timing cannot be established.
- Multiple observations are nested within users; the exploratory summaries do not model within-person correlation.
- Physical activity, sleep, stress, caffeine, alcohol, medication, illness, and unlogged food intake may confound individual events.
- The exact algorithm version and watch firmware/hardware version should be documented in the final submission for reproducibility.
- The exploratory association between meal composition and excursion is weak; this dataset does not establish proportionality of spike magnitude to carbohydrate or glycaemic load.
- Temporal alignment is a material methodological consideration in this Samsung Galaxy Watch dataset. The prespecified 30-minute shift changed meal-event classification substantially, which is reported transparently and limits claims about exact meal-response timing while leaving the meal-independent bidirectional transition result unaffected.

Clinical and Regulatory Interpretation

The present retrospective analysis supports balanced physiological directional responsiveness, heterogeneous meal-associated metabolic dynamics, post-peak recovery, and orthogonal confirmation that the study population exhibited expected fasting-to-postprandial state transitions. It should not be interpreted as numerical blood-glucose measurement, a substitute for CGM/glucometer testing, or evidence supporting therapeutic dosing decisions.

Conclusion

In de-identified real-world data from 28 Samsung Galaxy Watch users operating with the Agatsa One app, the proprietary sugar-trend algorithm produced a strongly bidirectional signal with closely matched frequencies of material rises and falls. Meal-linked analyses demonstrated heterogeneous event-associated dynamics and clear dependence on temporal alignment, while 63.9% of evaluable post-peak trajectories subsequently declined. Independently, after excluding the Calibration sheet, all 10 participants with clean manual/photo fasting and post-meal reference measurements showed the expected higher post-meal state. These convergent findings support physiological-anchor concordance as a useful feasibility framework for wearable-derived metabolic trend inference. They do not establish numerical blood-glucose accuracy or equivalence to a biochemical glucose measurement.

Declarations

Ethics approval and consent to participate. This manuscript reports a retrospective secondary analysis of de-identified real-world records generated during ordinary commercial use of the Agatsa One application and

related products. No experimental intervention, study-specific recruitment, participant payment, or study-mandated procedure was performed for this analysis. Before journal submission, the authors should document the applicable institutional ethics/IRB determination (approval, waiver, or exemption, as applicable under the relevant jurisdiction) and insert the committee name and reference number if one was issued.

Consent for publication. No directly identifying participant information, photographs, case narratives, names, telephone numbers, or other identifying details are included in the manuscript. Individual consent for publication of identifiable material is therefore not applicable to the manuscript as written.

Data availability. The analyses are based on de-identified event-level Samsung Galaxy Watch observations, time-stamped meal logs, and a limited non-calibration reference-sugar stream. The Calibration sheet was excluded from the orthogonal reference analysis. A de-identified minimum dataset sufficient to reproduce the reported aggregate analyses may be made available subject to participant privacy, contractual restrictions, and applicable data-protection requirements. The final journal submission should state the exact repository or controlled-access procedure selected by the authors.

Code and algorithm availability. The Agatsa One sugar-trend algorithm is proprietary software. The analysis definitions required to reproduce the statistical summaries in this manuscript are described in the Methods and Supplementary Appendix. The executable production algorithm and source code are not publicly released because they contain commercially confidential intellectual property. For peer review, sufficient technical documentation, version-locked outputs and analysis code should be made available to editors and reviewers under appropriate confidentiality arrangements where required by journal policy.

Funding. No external or internal study funding was provided for this retrospective analysis. Agatsa Software Pvt. Ltd. did not pay participants, provide study stipends, subsidize participation, or purchase devices for research use. The underlying records arose from ordinary customers who independently purchased Agatsa One subscriptions and/or EasyTouch Rhythm Band products with their own funds. Agatsa is the developer/operator of the commercial platform and proprietary algorithm and provided access to the de-identified product-use data used for this analysis; that role is disclosed separately as a competing interest.

Competing interests. Agatsa One, EasyTouch Rhythm Band, and the sugar-trend algorithm evaluated in this manuscript are commercial/proprietary technologies of Agatsa Software Pvt. Ltd. Authors who are founders, directors, employees, shareholders, consultants, or inventors associated with Agatsa must disclose those relationships explicitly in the final submission. These commercial interests do not constitute study funding; they are reported so readers can assess potential conflicts of interest.

Intellectual property. The manuscript describes a proprietary wearable-derived sugar-trend algorithm and a physiological-anchor concordance framework for evaluating directional metabolic responsiveness. Publication of this manuscript does not by itself determine inventorship, patentability or ownership. Any patent applications, issued patents or pending intellectual-property claims that are materially related to the algorithm or validation framework should be disclosed to the journal as competing interests where applicable and coordinated with publication timing.

Author contributions. Authorship and CRediT roles must reflect actual contributions. The final submission should identify, by author initials, contributions to Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Writing - original draft; Writing - review & editing; Visualization; Supervision; and Project administration, as applicable. No Funding acquisition role should be assigned for this study because no study funding was provided by Agatsa Software Pvt. Ltd.; any unrelated external support should be disclosed only if it actually contributed to this work.

Acknowledgements. The authors acknowledge the engineering and product teams responsible for operation of the Agatsa One platform and the customers whose de-identified real-world records contributed to this retrospective analysis. Customers were not recruited or compensated for participation in a research study.

Algorithm and device versioning. The manuscript concerns the Agatsa One sugar-trend algorithm operating with Samsung Galaxy Watch-derived data. Exact Agatsa One application build, production algorithm version/hash, compatible watch model(s), watch firmware version(s), server-side processing version and analysis-script version should be frozen at database lock and reported in the final submission. These identifiers were not contained in the supplied analytical workbook and are therefore not fabricated in this draft.

Reporting transparency. All analyses in the present manuscript are retrospective analyses of real-world product-use data. The 30-minute timing sensitivity analysis is reported explicitly because temporal alignment materially affects event anchoring. The orthogonal reference analysis excludes the Calibration sheet and is presented only as physiological-state context, not direct glucose validation. No prospective CGM protocol or future-study proposal is part of the present manuscript.

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Supplementary Appendix: Analysis Definitions Used in This Draft

This appendix is included so that the analytical claims in the manuscript are auditable and can be replaced by a locked statistical analysis plan before external submission.

Stored trend value: Midpoint of the source workbook sugar low-high interval.

Raw timestamp primary analysis: Meal time treated as $t=0$ because the workbook specifies that no lag has been applied to stored timestamps.

Timing tolerance: ± 30 minutes where required because median watch sampling interval is 30 minutes.

Post-meal evaluability: At least one pre-meal value in the baseline window and at least one value in the specified post-meal response window.

Material consecutive transition: Absolute change >3 units between consecutive observations separated by 15-90 minutes.

Sensitivity analysis: A 30-minute temporal shift was applied to quantify dependence on potential physiological/algorithmic latency.

No clinical-accuracy endpoint: No MARD, error-grid, or reference-glucose sensitivity/specificity is reported because no contemporaneous independent reference-glucose series was paired to the wearable observations. The separate non-calibration reference analysis tests only fasting-to-postprandial physiological-state ordering.

Orthogonal reference analysis: The workbook Calibration sheet and all records flagged as calibration references were excluded. The clean reference subset used manual/photo-captured non-calibration observations.

Participant-level medians compared fasting with post-meal states; a supportive same-day analysis compared fasting observations with later post-meal observations on the same date.